

Lesson 4 Buzzer Control

This section demonstrates how to control the buzzer on the expansion board to make sounds.

1. Working Principle

Set the “set_buzzer()” function to adjust the buzzer’s high and low voltage levels.

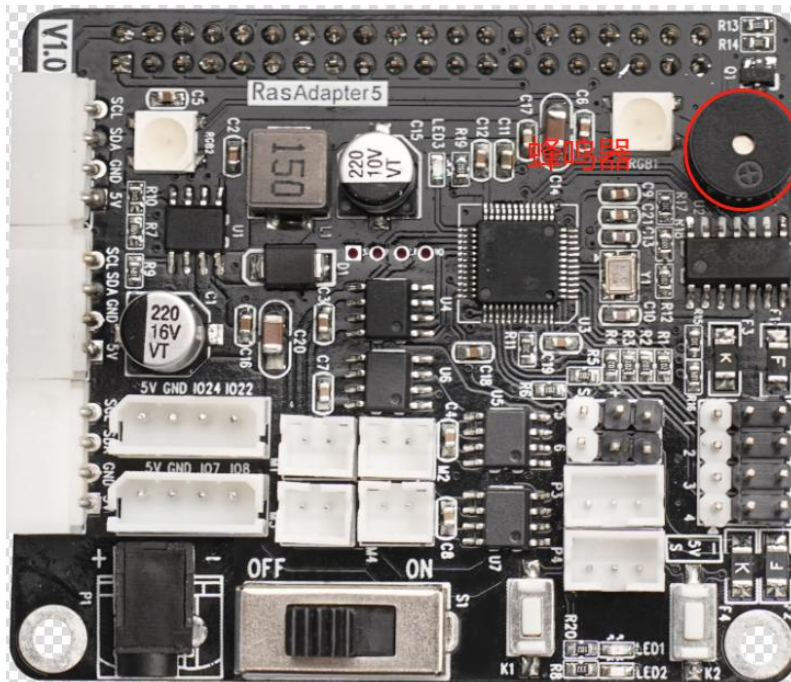
The source code of the program is located in “/home/pi/board_demo/buzzer_control_demo.py”.

```
4 print('')
5 *****
6 *****功能:幻尔科技树莓派扩展板, 蜂鸣器控制例程(function:Hiwonder Raspberry Pi
  expansion board, buzzer control)*****
7 *****
8 -----
9 Official website:https://www.hiwonder.com
10 Online mall:https://hiwonder.tmall.com
11 -----
12 Tips:
13 * 按下Ctrl+C可关闭此次程序运行, 若失败请多次尝试!(Press "Ctrl+C" to exit the p
  rogram. If it fails, please try again multiple times!)
14 -----
15 '')
16 board = rrc.Board()
17 board.set_buzzer(1900, 0.1, 0.9, 1) # 以1900Hz的频率, 持续响0.1秒, 关闭0.9秒, 重
  复1次(The buzzer sounds at a frequency of 1900Hz for 0.1 seconds followed by a p
  ause of 0.9 seconds, and repeat this pattern once)
18 time.sleep(2)
19 board.set_buzzer(1000, 0.5, 0.5, 0) # 以1000Hz的频率, 持续响0.5秒, 关闭0.5秒, 一
  直重复(The buzzer sounds at a frequency of 1000Hz for 0.5 seconds followed by a
  pause of 0.5 seconds, and repeat this pattern once)
20 time.sleep(3)
21 board.set_buzzer(1000, 0.0, 0.0, 1) # 关闭(close)
```

The program calls the “set_buzzer()” function in the “Board” library to adjust the buzzer’s high and low voltage levels. The parameters in the “set_buzzer(1900, 0.1, 0.9, 1)” function are set to a frequency of 1900Hz, with the buzzer sounding for 0.1s, followed by a 0.9s pause, and repeating once.

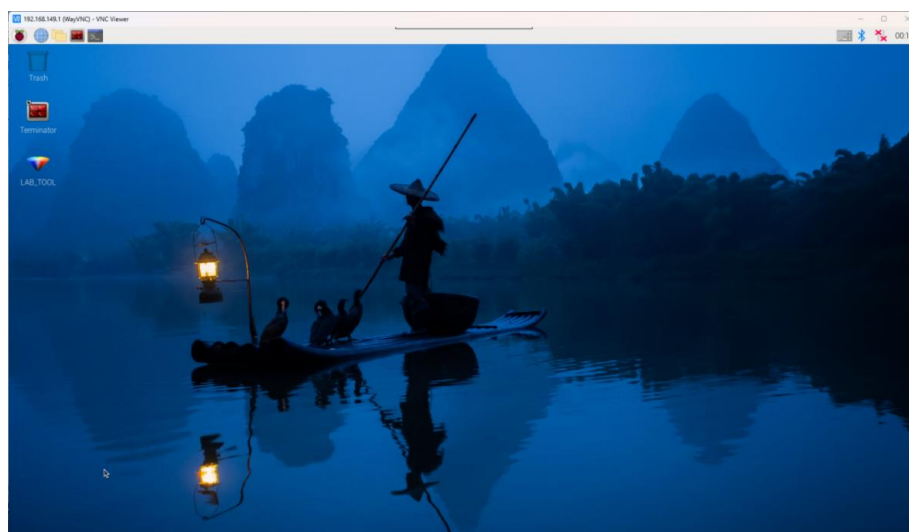
2. Getting Ready

A buzzer is mounted on the Raspberry Pi expansion board, as shown below:

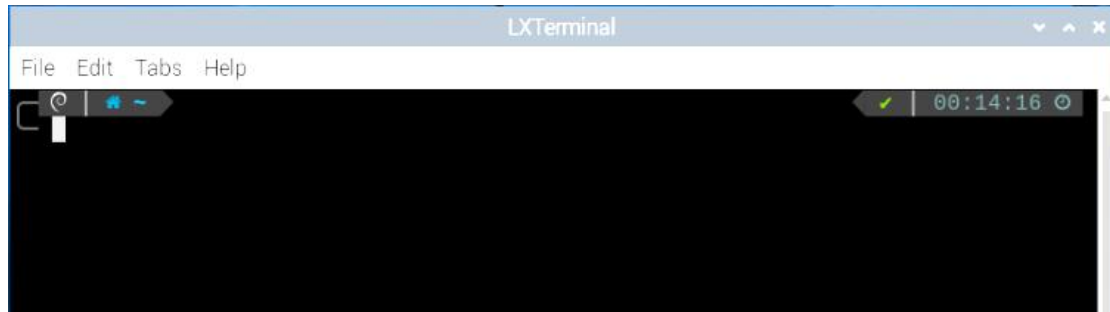


3. Operation Steps

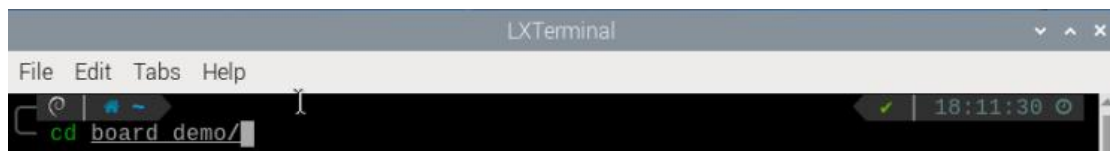
1) Turn the device on and connect the robot through the VNC remote connection tool.



2) Click  in the top left corner of the desktop, or press the shortcut key “Ctrl+Alt+T” to open the command line terminal.



3) Enter the command “cd board_demo/” in the directory where the program is, and press “Enter”.



4) Enter the command “python3 buzzer_control_demo.py”, and press “Enter” to activate the game. The buzzer will make a short sound and then three long sounds to automatically exit the program.



4. Program Outcome

After the program is activated, the buzzer on the Raspberry Pi expansion board makes a short sound and three long sounds to exit the program.